HONGZE YU

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EDUCATION

University of Michigan, Ann Arbor, USA *Aug. 2023 – Present*
Ph.D. Candidate in Electrical and Computer Engineering (GPA: 4.00/4.00)
Advisors: Prof. Yun Jiang and Prof. Jeffrey A. Fessler

University of Michigan, Ann Arbor, USA *Aug. 2021 – Apr. 2023*
M.S. in Electrical and Computer Engineering (GPA: 4.00/4.00)

University of Glasgow, UK *Sep. 2019 – Jun. 2021*
B.E. in Electronics and Electrical Engineering, First Class Honours (GPA: 19/22)

University of Electronic Science and Technology of China, China *Sep. 2017 – Jun. 2019*
B.S. in Electronic Information Engineering, 2+2 Joint Program (GPA: 3.97/4.00)

RESEARCH INTERESTS

My research focuses on developing mathematical and computational methods for medical imaging, with the goal of improving image quality, reducing MRI scan time, and supporting clinical patient care.

RESEARCH PROJECTS

Joint Coordinate-Based Representation for Fast Magnetic Resonance Fingerprinting
Aug. 2025 – Present

- Developed a coordinate-based continuous representation method for magnetic resonance fingerprinting (MRF) reconstruction, a quantitative MRI technique used to estimate tissue parameters such as T_1 and T_2.
- Improved consistency across reconstructed images to reduce artifacts and improve the quality of quantitative tissue maps.
- Evaluated the method using phantom and brain MRI data and compared the resulting T_1/T_2 maps with standard reconstruction approaches.
- Demonstrated improved image quality and reduced reconstruction time for quantitative MRI.

Patch-Based Diffusion Reconstruction for Prostate MRI
Aug. 2025 – Present

- Developed a patch-based reconstruction method for accelerated prostate T_2-weighted MRI.
- Applied the method to prostate MRI data acquired at standard clinical and low-field scanner settings.
- Improved image quality by reducing noise and reconstruction artifacts compared with standard reconstruction methods.
- Collaborated with radiologists on reader studies to assess reconstructed image quality.

Bilevel-Optimized Coordinate-Based Representation for Accelerated MRI Reconstruction
Jan. 2024 – Feb. 2026

- Developed a coordinate-based continuous representation method to improve image quality from accelerated MRI scans.
- Designed an automatic parameter-selection approach using only the acquired MRI data.
- Evaluated the method across multiple anatomies, scanner settings, and sampling patterns.

· Demonstrated reduced image artifacts and improved image-quality metrics compared with conventional reconstruction methods.

PUBLICATIONS

Journal Article

[1] **H. Yu**, J. A. Fessler, Y. Jiang, "Bilevel Optimized Implicit Neural Representation for Scan-Specific Accelerated MRI Reconstruction," *IEEE Transactions on Medical Imaging*, 2026. [doi].

Preprint

[2] **H. Yu**, Y. Jiang, J. A. Fessler, "Comparing Implicit Neural Representations and B-Splines for Continuous Function Fitting from Sparse Samples," preprint, 2026. [arXiv].

CONFERENCES & WORKSHOPS

[1] **H. Yu**, J. Hu, M. Jaroszewicz, H. K. Hussain, V. Gulani, J. A. Fessler, Y. Jiang,
"Patch-Based Diffusion Inverse Solver for T2-Weighted Prostate Imaging Reconstruction."
ISMRM Workshop on Data Sampling and Image Reconstruction, Sedona, 2026. **(Traditional Poster)**
Accepted to the *2026 ISMRM Annual Meeting.* **(Digital Poster)**

[2] **H. Yu**, C. Keen, K. Jin, J. A. Fessler, Y. Jiang,
"Joint Implicit Neural Representation for Fast Scan-Specific Magnetic Resonance Fingerprinting."
ISMRM Workshop on Data Sampling and Image Reconstruction, Sedona, 2026. **(Traditional Poster)**
Accepted to the *2026 ISMRM Annual Meeting.* **(Oral Presentation)**

[3] **H. Yu**, J. A. Fessler, Y. Jiang,
"Bilevel Optimized Implicit Neural Representation for Scan-Specific Accelerated MRI Reconstruction."
33rd Annual Meeting of the ISMRM, Hawaii, 2025. **(Oral Presentation, Summa Cum Laude Award)**

SKILLS

Medical imaging, MRI reconstruction, image processing, signal processing, mathematical modeling, numerical optimization, Python, MATLAB, Julia

STUDENT MENTORING

Kaixuan Jin, undergraduate research project on MRI reconstruction, University of Michigan *Jul. 2025 – Dec. 2025*

REFERENCES

Yun Jiang [Homepage]
Jeffrey A. Fessler [Homepage]